

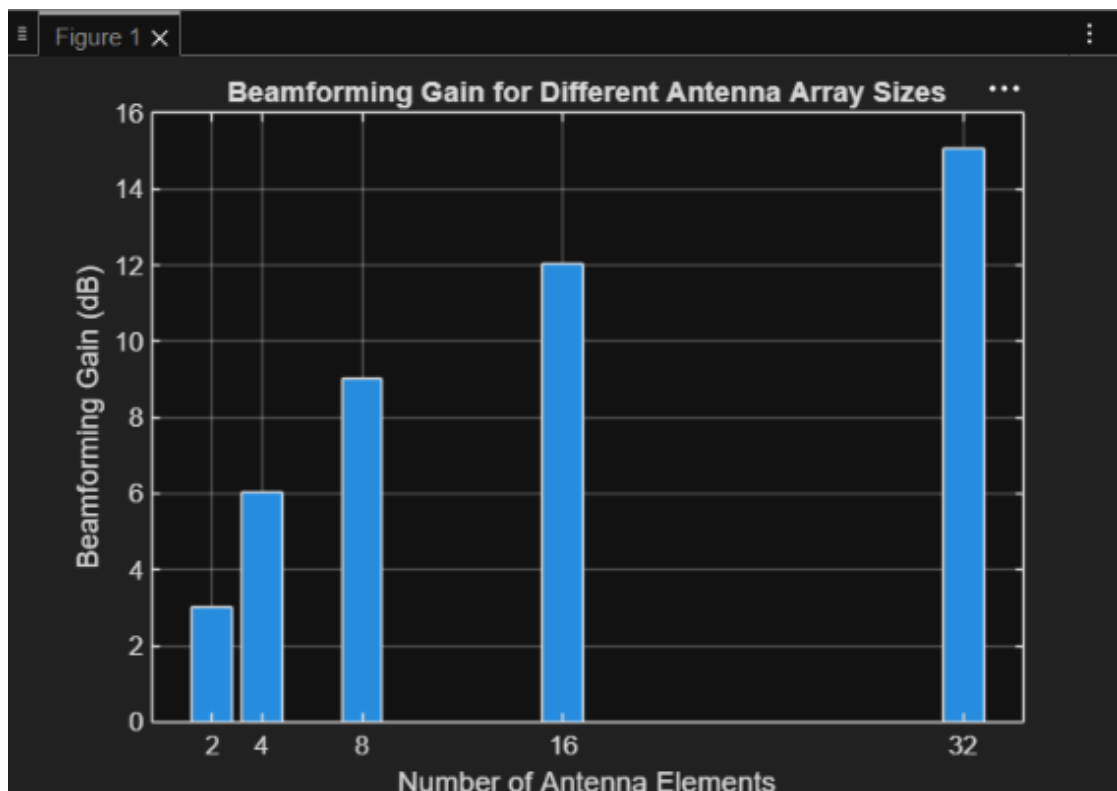
EXP :7 Analog Beamforming Analysis Using MATLAB

Objective – 1: Analyze Beamforming Gain (dB)

Program code:

```
clc;
clear;
close all;
antennas = [2 4 8 16 32]; % Number of Antenna Elements
gain = 10*log10(antennas); % Beamforming Gain (dB)
figure
bar(antennas, gain)
grid on
xlabel('Number of Antenna Elements')
ylabel('Beamforming Gain (dB)')
title('Beamforming Gain for Different Antenna Array Sizes')
```

Output:

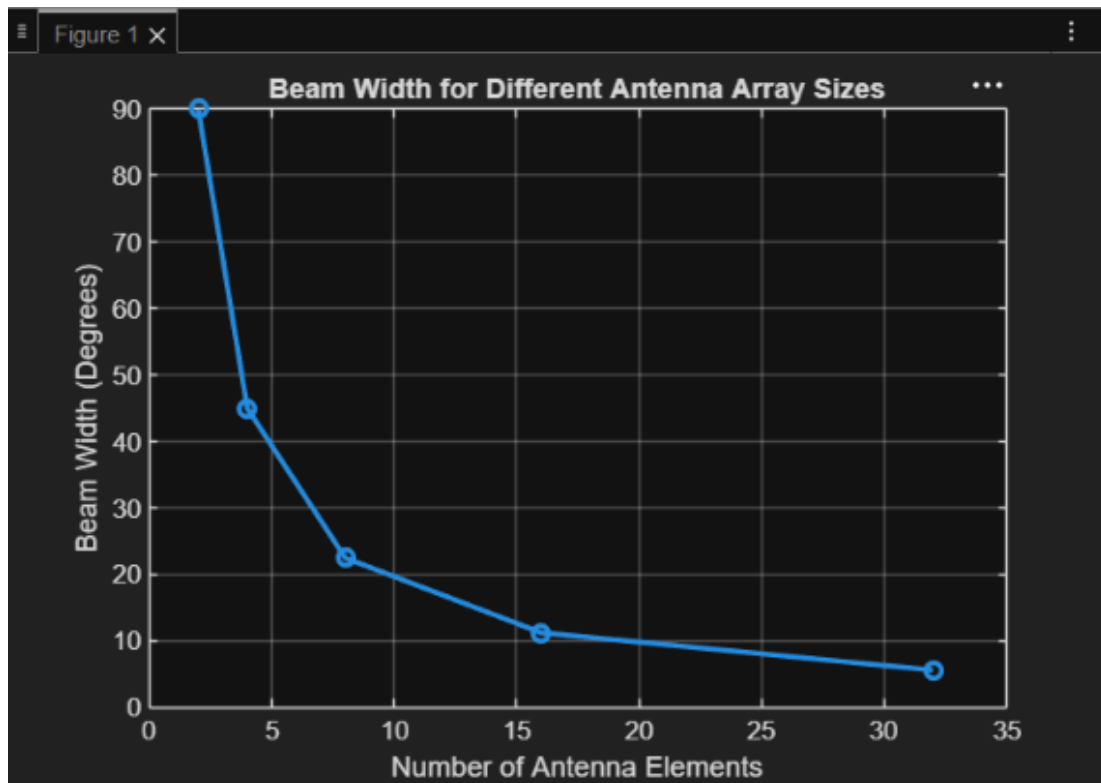


Objective – 2: Compare Beam Width (Degrees)

Program code:

```
clc;
clear;
close all;
antennas = [2 4 8 16 32]; % Number of Antenna Elements
beamwidth = [90 45 22.5 11.25 5.625]; % Beam Width (degrees)
figure
plot(antennas, beamwidth, 'o-', 'LineWidth', 2)
grid on
xlabel('Number of Antenna Elements')
ylabel('Beam Width (Degrees)')
title('Beam Width for Different Antenna Array Sizes')
```

Output:



Objective – 3: Analyze Main Lobe Direction (Degrees)

program code:

```
clc;
clear;
close all;
theta = [-60 -30 0 30 60]; % Main Lobe Direction (degrees)
gain = [1 1 1 1 1]; % Constant normalized gain
figure
polarplot(deg2rad(theta), gain, 'o', 'LineWidth', 2)
title('Main Lobe Direction in Analog Beamforming')
```

Output:

